

Electroencephalography Data Denoising with Deep Neural Networks

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Abstract—Electroencephalography (EEG) is frequently utilized in investigations including neuroscience, neural engineering, and biomedical engineering, because of its benefits, such as non-invasiveness, high temporal resolution, and relatively low cost. Nevertheless, the raw EEG data are frequently corrupted by physiological artifacts and many noises, such as cardiac artifacts, myogenic artifacts, and ocular artifacts. Artifacts can have a negative impact on an EEG signal and can be reduced or eliminated using existing deep learning-based EEG signal denoising techniques. However, when the acquired EEG data has significant artifacts, they typically experience performance degradation. In order to address this problem, we introduced the 2x3R-CNN and compared it to four deep learning-based methods: fully connected neural network (FCNN), basic convolution network, and deep reinforcement learning (simple CNN), complex convolution network (complex CNN), and recurrent neural network (RNN) to remove muscle and ocular artifacts from EEG signal. The experimental results demonstrate that, on the EEGdenoiseNet dataset, the proposed model outperforms the existing approaches.

Keywords—convolutional neural network, EEG denoising, electroencephalography

I. INTRODUCTION

The use of electroencephalography (EEG) for brain signal analysis has been broadly practiced since its creation, highly advancing the neurological and medical industries. Several physiological noises, including cardiac artifacts, myogenic artifacts, and ocular artifacts, typically contaminate the raw EEG data. A significant challenge that has hampered the accurateness of EEG signal analysis is the matter of noise. Therefore, an essential track to empower the research of neural activity is EEG data denoising [1]. Numerous methods have been undertaken with the aim of removing noisy signals from EEG data. These approaches include many conventional denoising methods [2-4]. The intrinsic structure of neuronal fluctuations in EEG can be discovered using DL-based approaches [5-7], which can then be used to reduce

oscillations that are caused by biological noises rather than genuine brain processes.

This study proposed a novel EEG data denoising model with a deep neural network called 2x3R-CNN which can successfully reconstruct the denoised EEG data through a single fully connected layer and multiple convolutional layers. This model addresses the generalization issue of the two methods belong to the CNN-based methods [8, 9] and can yield cleaner EEG data and achieve significant improvement. This work hypothesized that the amount of feature maps may be what restricts the generalization ability of the two approaches [8, 9] by comparing their composition to that of VGGNet [10], U-net [11], and ResNet [3], which does not experience generalization problem. This study, which was inspired by U-net, VGGNet, and ResNet, increased the number of feature maps and downsampled the data in a temporal domain in order to reduce the amount of features and avoid overfitting [12]. Interestingly, 2x3R-CNN not only solved the generalization problem but also demonstrated a better performance.

The following are the study's main contributions:

- 1) In order to train a network deeper with better artifact removal capabilities, we developed a 2x3R-CNN to remove artifacts from EEG signal in which skip-layer using ResNet connections is presented to avoid gradient detonation [8]. The multi-scale characteristics were eliminated by this model by repeatedly stacking the remaining blocks.
- 2) By calculating relative root mean square error in the spectral field and temporal field, and the average correlation coefficient, we have covered comparisons between the performance of 2x3R-CNN and other networks.
- 3) The source code of proposed model can be found at <https://github.com/wondimagegn-b/EEG-Denoising>

The article also includes the experimental findings that demonstrate how the suggested model can provide denoised EEG data, improve denoising performance, and increase generalizability.

II. MATERIALS AND METHODS

A. Data Collection

The EEGdenoiseNet benchmark dataset [8] is the source of the EEG, electrooculography (EOG), and electromyography (EMG) data used in this study. In this benchmark dataset, the data were cleaned, segmented into 2-second epochs, and measured to a similar variance before being visually inspected to ensure they were ready for usage.

The EEG dataset included 64 channels that were captured at 512 Hz using the international 10-10 standard for both real and fictitious motion situations in the left and right hands. The data were then resampled to 256 Hz, band-pass processed within 1 and 80 Hz, and ragged at powerline frequency. ICLabel [3] was then used to control the 64-channel signals in order to eliminate noisy data. The EEG data were then separated into 2 second epochs. The final dataset included 3400 ocular artifact epochs, 5598 myogenic artifact epochs, and 4514 EEG epochs.

B. Data Usage

By linearly integrating the EEG epochs, EMG epochs, and EOG epochs as shown in (1), and utilizing the signal to noise ratios (SNRs), as shown in (2), picked from a uniform range from -7dB to 2dB, the study randomly mixed artifact data and actual EEG data ten times for the test set and training set. In this way, the mixed epochs—which are thought to be contaminated EEG—and the pure EEG epochs might serve as the basis for comparison.

$$y = x + \lambda * n \quad (1)$$

$$SNR = 10 \log \frac{RMS(x)}{RMS(\lambda * n)} \quad (2)$$

In the calculations, y stands for mixed data of artifacts (ocular or myogenic) and EEG, x for actual EEG data, for artifact involvement relative to EEG, and n for artifacts.

C. Generation of Training and Test Dataset

We combined 3400 sets of pure EEG epochs and EOG epochs from the dataset [8] to produce ocular artifact-polluted data, using 400 pairs for the test dataset and 3000 pairs for the training dataset. We randomly combined 3000 pairs of pure EEG signals with ocular artifact signals 10 times for the training set, with the SNR value depending on fair distribution and ranging from -7dB to 2dB. The trained dataset was expanded in this way to 30000 epochs. Similarly, the testing dataset was enlarged to 4000 epochs by arbitrarily combining 400 sets of epochs from pure EEG data and ocular artifact data with ten different SNR levels.

We combined 4500 pure EEG epochs with EMG artifact epochs to create tainted signals caused by myogenic artifacts. 500 pairings were utilized to create the testing dataset, and a total of 4000 pairs were used to create the train dataset. By applying the SNR from -7 to 2dB, the amount of the train and

testing datasets rose ten times greater, similar to the procedure for ocular artifact-polluted signals. In this way, the training set and test sets were each expanded by 5000 epochs.

III. PROPOSED METHOD

In order to train a deeper network with enhanced artifact-removal capabilities, this article developed an unique convolutional neural network called 2x3R-CNN, as illustrated in fig. 1, to denoise EEG data in which skip-layer using ResNet connection is presented to avoid gradient detonation [3]. The quantity of hidden convolution neural layers to be selected will be applied with the aim of making an uncomplicated convolutional neural network, dropping calculation time but with sufficient complexity to learn universal features and local features. The new model is known as 2x3R-CNN because it organizes three groups of residue blocks stems side by side in order to eliminate multi-scale characteristics. This model stacks the residual blocks repeatedly using convolutional kernels (1,3), (1,5), and (1,7) of numerous scales twice. In each of these blocks, followed by a dropout regularization, two 1D-convolution layers, 1 stride, and a ReLU activation function were used [15]. As the goal of this DL-based method is to reproduce the denoised EEG data, an appropriate activation function must be applied. Certain activation functions, such as *ReLU*, produce values within a limited range. The final convolution layer was preceded with a dense and flatten layer with similar outputs and inputs to reconstruct the signal. The first layer's input of contaminated EEG data was followed by the last layer's discovery of cleaned EEG data. The epoch number at which the validation curve of the training loss and validation curve of test loss of the data converges will be selected as the perfect epoch number. The batch size to be applied will be large enough to avoid sample bias, but small enough in order not to overfit the model.

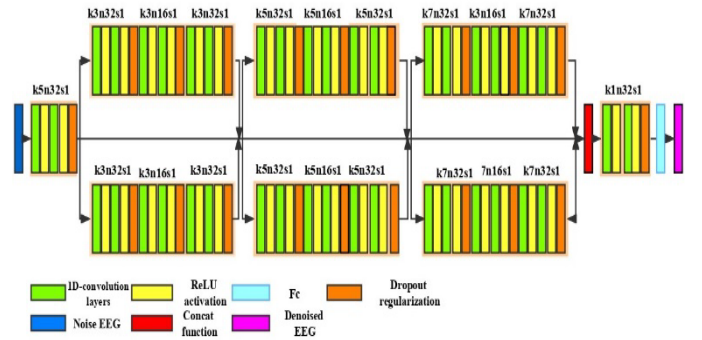


Fig.1. The architecture of the 2x3R-CNN

A. Learning Process

This study applied the concept from [8] and divided the standard deviation of raw EEG data to normalize the ground-truth EEG signal, x , and the input contaminated EEG signal, y , with the goal of smoothing the application and training of networks as shown in (3):

$$x = \frac{x}{\sigma_y}, \quad y = \frac{y}{\sigma_y} \quad (3)$$

here σ_y represents the contaminated data's standard deviation, which when multiplied by the correspondingly cleaned final EEG data restores the actual amplitude of the EEG. The input for neural networks will be y .

As seen in (4), the objective of a 2x3R-CNN model is to produce a nonlinear function $\tilde{z}$ that translates normalized raw EEG data y to the denoised EEG $\tilde{n}$:

$$\tilde{n} = \tilde{z}(y; \theta) \quad (4)$$

where $\tilde{n} \in R^{1 \times T}$ symbolizes the cleaned EEG data as the outcome, $\hat{y} \in R^{1 \times T}$ symbolizes the normalized contaminated data as the input, and θ is the collection of training parameters. The artifact removing model can also be expressed as a function $\tilde{z}$ that projects input data from $R(y)$ to a distribution $R(z)$, in which the divergence between $R(\tilde{n})$ and $R(z)$ is reduced. This assumes that $\tilde{n}$ is taken from the pure EEG data distribution $R(\tilde{n})$ and y as taken from the contaminated EEG data distribution $R(y)$. In the artifact removing model, the noisy EEG data $R(y)$ could become the denoised EEG data $R(z)$.

The adaptive moment estimation (Adam) approach was utilized to improve the loss function to optimize the 2x3R-CNN model. In contrast to stochastic gradient descent, Adam employs momentum and adaptive learning processes to converge more quickly and performs better. This helps us avoid becoming stuck in local minimums and saddle spots. Adam performs better than other adaptive techniques in practice. The PyTorch library was used to implement the model.

B. Performance Evaluation

RRMSE in the temporal field (RRMSEtemporal, as shown in (5)), RRMSE in the spectral field (RRMSEspectral, as shown in (6)), and the average correlation coefficient (ACC, as shown in (7)) are three objective measures used to analyze the performance of the proposed method on the denoised data [16].

$$RRMSE_{temporal} = \frac{RMS(f(y) - x)}{RMS(x)} \quad (5)$$

$$RRMSE_{spectral} = \frac{RMS(PSD(f(y)) - PSD(x))}{RMS(PSD(x))} \quad (6)$$

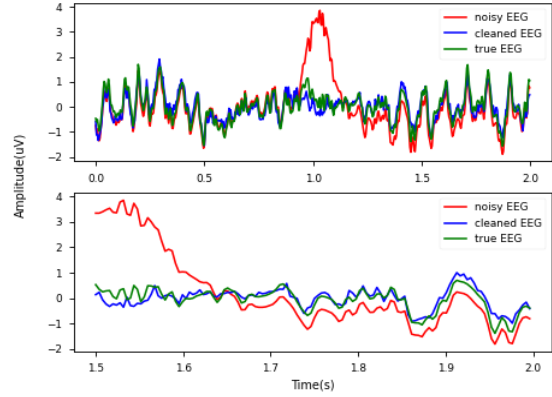
$$ACC = \frac{Cov(f(f(y)), x)}{\sqrt{Var(f(y))Var(x)}} \quad (7)$$

IV. EXPERIMENTAL RESULTS AND DISCUSSION

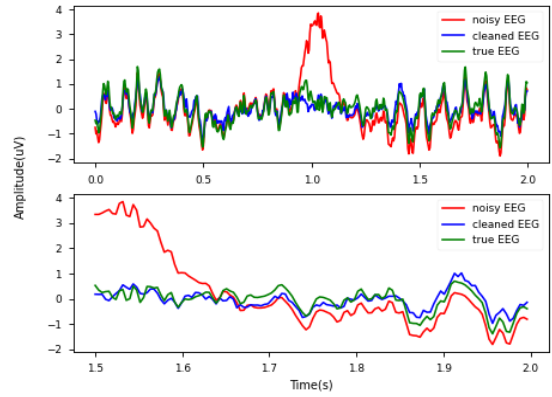
To describe a general outlook of the experimental result, the algorithms are evaluated in both qualitative and quantitative evaluation, by applying RRMSEtemporal, RRMSEspectral, and ACC. In fig.2, some instance epochs in terms of both time and frequency domains in the test process for ocular artifact removal are displayed, and in fig.3, myogenic artifact removal are displayed.

In general, a 2x3R-CNN model performed well in rebuilding the denoised EEG data samples, both in terms of ocular and myogenic artifact removal, compared to FCNN, RNN, basic CNN, and complex CNN. The frequency domain chart shows that the low frequency artifacts were clearly identified and removed. As an example, the denoised signal is much closer to the genuine signal when compared to contaminated and denoised signals acquired from a 2x3R-

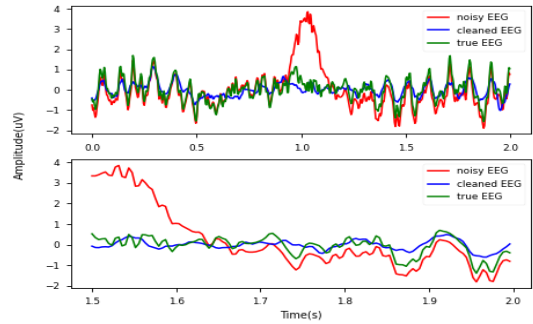
CNN. The 2x3R-CNN model denoises EEG data with excellent accuracy, and the denoised signals seem extremely near to the genuine data having an error near to 0, as illustrated in both Figs. 2 (ocular artifact removal) and 3 (myogenic artifact removal).



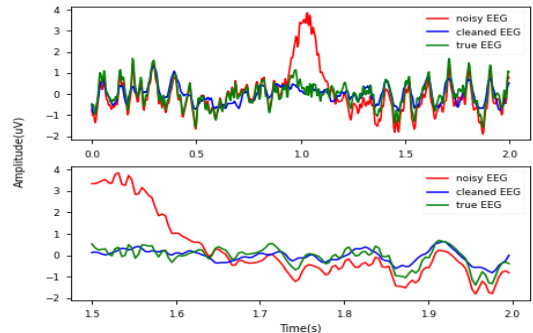
(a) Simple CNN



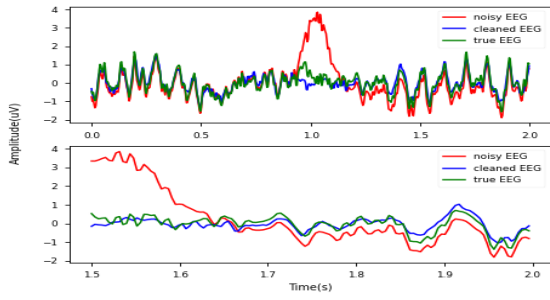
(b) Complex CNN



(c) FCNN

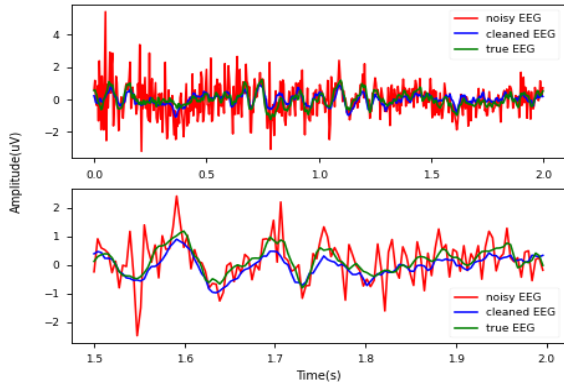


(d) RNN

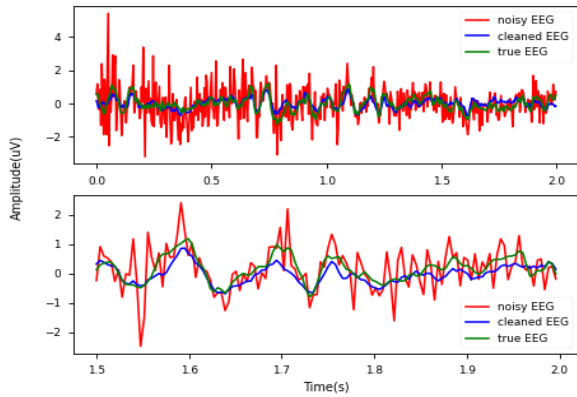


(e) Proposed network (2x3R-CNN)

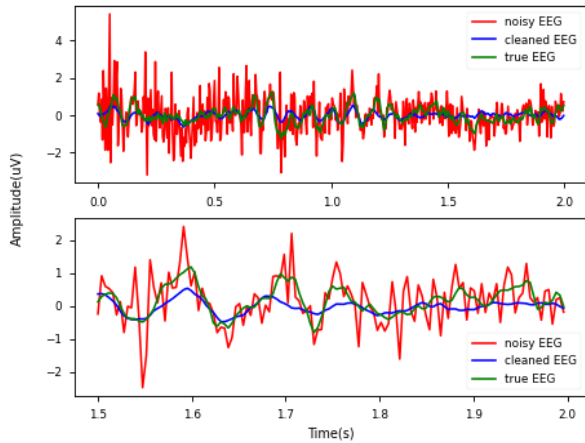
Fig.2. Some exemplary epochs of the performance for ocular artifact removal.



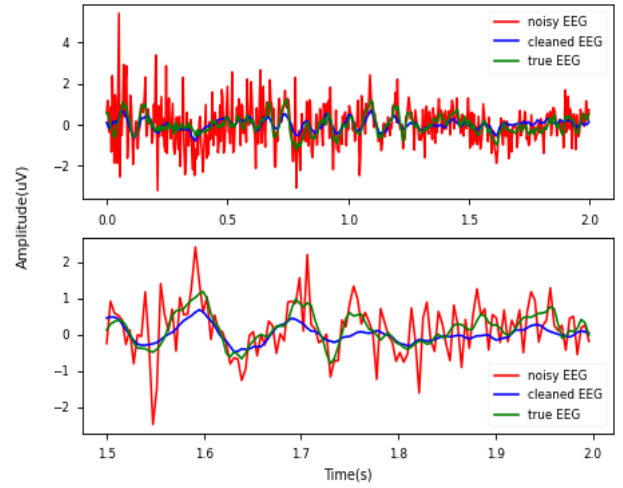
(a) Simple CNN



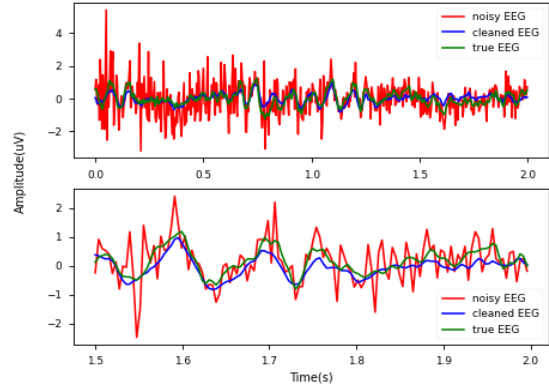
(b) Complex CNN



(c) FCNN



(d) RNN



(e) Proposed network (2x3R-CNN)

Fig.3. Some exemplary epochs of the performance for myogenic artifact removal.

In addition, this study averaged the SNR values for all ocular artifact removal, as shown in Table I, and myogenic artifact removal, as shown in Table II, to summarize the performance of the networks. The 2x3R-CNN had the lowest average $RRMSE_{temporal}$, $RRMSE_{spectral}$, and highest average ACC, as could be shown through ocular artifact removal and myogenic artifact removal. The values of the three quantitative metrics can be seen in Tables I (ocular noise removal) and II (myogenic noise removal), demonstrating that the efficiency of the proposed approach (2x3R-CNN) is superior to FCNN, RNN, and the two methods that are CNN-based methods.

TABLE I. OCULAR ARTIFACT REMOVAL SNRS

Model	$RRMSE_{temporal}$	$RRMSE_{spectral}$	ACC
Simple CNN	0.620	0.526	0.791
Complex CNN	0.630	0.588	0.776
FCNN	0.803	0.713	0.590
RNN	0.740	0.696	0.677
Proposed	0.552	0.512	0.832

TABLE II. MYOGENIC ARTIFACT REMOVAL SNRS

Model	$RRMSE_{temporal}$	$RRMSE_{spectral}$	ACC
Simple CNN	0.750	0.697	0.706
Complex CNN	0.746	0.680	0.692
FCNN	0.827	0.781	0.580
RNN	0.785	0.775	0.636
Proposed	0.721	0.669	0.707

V. CONCLUSION

In this study, A DL-based approach to remove ocular and myogenic artifacts from EEG data has been used. In order to achieve these goals, a 2x3R-CNN model has been designed and proposed using a convolutional neural network framework, with the goal of eliminating the noise existing in the EEG data, while keeping the important signal information. The anticipated result of the research was to eliminate the noise of the EEG data, with better performance. With a combination of numerous convolutional layers and a single fully connected layer, the proposed model can rebuild the cleaned EEG data. Several performance evaluations were used to examine the effectiveness of 2x3R-CNN, fully connected neural network, recurrent neural network, and the two CNN-based approaches to remove muscular and ocular artifacts from EEG data. According to experimental findings, 2x3R-CNN offers a lot of potential for removing ocular and myogenic noises from EEG data, even at high noise levels.

This work only focuses on removing ocular and myogenic artifacts from 2s-long EEG epochs that have been denoised. In practical applications, a better design is needed to denoise EEG epochs longer than 2 seconds and to better distinguish the characteristics of neural impulses from various electrical aberrations. To address this issue, 2x3R-CNN will be changed in the future. An additional network block to the existing networks that allows for the learning of the concealed relationship between continuous epochs for the removal of various electrical artifacts is one potential solution. Moreover, DL networks can be applied to different tasks including feature extraction and EEG categorization.

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